Emanuele Rossi

emanuele.rossi1909@gmail.com Website Scholar LinkedIn GitHub

Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously ML on graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,444 citations; h-index 15 (August 2026).

Education

Imperial College London *Oct 2020 – Feb 2024*
Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge *Oct 2018 – Jun 2019*
Master in Advanced Computer Science, Distinction (**4.0 GPA**)

Imperial College London *Oct 2015 – Jun 2018*
Bachelor in Computer Science, First Class Honors (**4.0 GPA**)

Experience

Postdoctoral Researcher *Rome (remote)*
GLADIA Research Group, Sapienza University of Rome – With Prof. E. Rodolà *Oct 2025 – Present*

- Developing machine learning methods for animal communication.

Machine Learning Researcher *NYC (remote)*
VantAI *Jan 2024 – Aug 2025*

- **Co-led development of Neo-1**, a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher *London*
Twitter *Jun 2019 – Feb 2023*

- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher *London*
Fabula AI *Mar 2019 – Jun 2019*

- Developed graph ML models for fake news detection as part of Fabula AI's small research team before Twitter acquired the company

Software Engineering Intern *California*
Google *Jul 2017 – Oct 2017*

Selected Publications

Full list

Animal Communication & Audio

- *D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and **E. Rossi**.* Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%).*
- *F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi** et al.* OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review 2026.*
- ***E. Rossi** and E. Rodolà.* Communicating Sound Through Natural Language. *arXiv 2026.*

Graph Learning

- ***E. Rossi** et al.* Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML GRL*

Workshop 2020. **1,700+ citations** and **1,000+ GitHub stars**.

- ***E. Rossi****, *F. Frasca** *et al.* SIGN: Scalable Inception Graph Neural Networks. *ICML GRL Workshop 2020.* **600+ citations** and used in **production at Airbnb**.
- *B. Chamberlain, S. Shirobokov,* ***E. Rossi*** *et al.* Graph Neural Networks for Link Prediction with Subgraph Sketching. *ICLR 2023, Oral (top 5%).* **200+ citations**.

Leadership

Co-Founder, LeadTheFuture (Forbes article) *Jun 2018 – Mar 2023*

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**

Supervision and Mentoring

- **Davide Marincione** – PhD Student @ Sapienza University of Rome *05/2025 – 03/2026*
  Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models
- **Simone Antonelli** – ML Research Intern @ Amboss Technologies *10/2024 – 07/2026*
  Predicting Link Closure in the Lightning Network with Temporal Graph Networks
- **Marco Pegoraro** – ML Research Intern @ VantAI *10/2024 – 04/2025*
  Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Harrison Rush** – ML Engineer @ Amboss Technologies *08/2024 – Present*
  MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning
- **Thomas Castiglione** – ML Researcher @ VantAI *03/2024 – 04/2025*
  Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules
- **Andy Huang** – PhD Student @ Mila *03/2024 – 04/2025*
  TGB: Temporal Graph Benchmark for Machine Learning on Temporal Graphs
  UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs
- **Vincent Davis** – ML Engineer @ Amboss Technologies *11/2023 – 05/2024*
  Channel Balance Interpolation in the Lightning Network via Machine Learning

Research Grants & Fellowships

- **ISCRA-C Research Computing Grant** – CINECA *02/07/2026*
  DOLPH-ID: Dolphin Emitter Identification from Whistle Acoustics **≈€20,000** equivalent
  Allocation of 8,500 GPU-hours on Leonardo's NVIDIA A100 64 GB accelerators. 9 months
- **Research Fellowship** – Sapienza University of Rome *01/04/2026*
  For postdoctoral research on machine learning for non-human communication. **€36,000**
  12 months
- **Google Cloud Research Credits** – Google *2022, 2023*
  GCP credits for PhD research. Received in two consecutive years. **$1,000/year**

Awards

- **Olav Beckmann Project Prize** – Imperial College London *2017*
  For outstanding second-year project work.
- **Engineering Dean's List** – Imperial College London *2016, 2017*
  Top 10% of cohort. Received in two consecutive years.
- **G-Research Prize (Gloucester Research Ltd Prize at the time of award)** – Imperial College London *2016*
  Awarded annually to up to 10 non-final-year Computer Science students for academic excellence.

- **President's Undergraduate Scholarship** – Imperial College London *22/05/2015*
  Imperial's most prestigious undergraduate distinction, awarded to fewer than 5% of the incoming cohort for academic excellence and potential. Funding: £3,000.

Invited Talks

- *Graph Machine Learning for Dynamic Financial Networks: From Foundations to the Lightning Network* *10/07/2026*
  Swissquote Workshop on Graph Learning in Financial Networks, EPFL [Slides (HTML)] [Slides (PDF)]
- *Machine Learning on Temporal Graphs: From Foundations to Emerging Frontiers* *19/05/2026*
  Linguae-ML Seminar, Ecole Normale Superieure (Paris) [Slides (HTML)] [Slides (PDF)]
- *Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models* *30/01/2026*
  Earth Species Project Community Deep Dive [Slides (PDF)] [Video]
- *A Primer on Graph Neural Networks* *22/11/2023*
  COLT Seminar, University of Pompeu Fabra, Barcelona [Slides (PDF)]
- *Edge Directionality Improves Learning on Heterophilic Graphs* *14/09/2023*
  Temporal Graph Reading Group [Slides (PDF)] [Video]
- *Edge Directionality Improves Learning on Heterophilic Graphs* *07/09/2023*
  Explainability and Applicability of GNNs Workshop, Kassel
- *Improving Machine Learning at Twitter using Graphs* *28/09/2022*
  Privacy Week [Video]
- *Learning on Graphs with Missing Node Features* *09/09/2022*
  TigerGraph Reading Group
- *Learning on Graphs with Missing Node Features* *01/03/2022*
  Learning on Graphs and Geometry Reading Group [Slides (PDF)] [Video]
- *Learning on Graphs with Missing Node Features* *24/02/2022*
  Machine Learning Research Group, University of Oxford
- *Learning on Graphs with Missing Node Features* *23/11/2021*
  University of Cambridge AI Research Group Talks
- *Graph Neural Networks with Almost No Features* *17/11/2021*
  Toronto Machine Learning Summit
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* *16/06/2021*
  MLOps World 2021
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* *26/04/2021*
  Intel ML Seminar [Slides (PDF)]
- *Machine Learning on Dynamic Graphs: Temporal Graph Networks* *09/04/2021*
  GNNSys'21 Workshop
- *Dynamic Graphs and Temporal Graph Networks* *22/01/2021*
  University of Liverpool Networks and Distributed Computing Series [Slides (PDF)]
- *TGN: Temporal Graph Networks* *21/07/2020*
  AISC [Slides (PDF)] [Video]

Academic Service

- **Post-AGI Science and Society Workshop** – ICLR 2026 *04/2026*
  PC Member
- **Temporal Graph Learning Workshop** – NeurIPS 2023 *12/2023*
  Organizer
- **Temporal Graph Learning Workshop** – NeurIPS 2022 *12/2022*
  Panel

- **NeurReps Workshop – Symmetry and Geometry in Neural Representations** – NeurIPS 2022 *12/2022*
  PC Member
- **GCLR 22 – 2nd Workshop on Graphs and more complex structures for learning and reasoning** – AAAI 2022 *02/2022*
  PC Member
- **GReS – Workshop on Graph Neural Networks for Recommendation and Search** – ACM RecSys 2021 *10/2021*
  PC Member
- **GNNSys'21 – Workshop on Graph Neural Networks and Systems** – MLSys 2021 *04/2021*
  PC Member
- **AAAI-21 GCLR** – AAAI 2021 *02/2021*
  PC Member

Full Publication List

Under Review

- F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi**, Y. Lakretz, G. G. de Polavieja and G. Sumbre. 2026. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review.* (**0**).
- H. Rush, V. Davis, S. Antonelli, V. Singh, J. Shrader and **E. Rossi**. 2026. MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning. *arXiv.* (**0**).
- **E. Rossi** and E. Rodolà. 2026. Communicating Sound Through Natural Language. *arXiv.* (**0**).
- S. Antonelli, V. Davis, H. Rush, A. Potdevin, J. Shrader, V. Singh and **E. Rossi**. 2026. Predicting Channel Closures in the Lightning Network with Machine Learning. *arXiv.* (**0**).
- D. Kovtun, M. Akdel, A. Goncearenco, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, **E. Rossi**, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein and L. Naef. 2024. PINDER: The Protein Interaction Dataset and Evaluation Resource. *bioRxiv.* (**37**).

Peer-reviewed journal articles

- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2026. Training-Free Graph-Based Imputation of Missing Modalities in Multimodal Recommendation. *IEEE Transactions on Knowledge and Data Engineering.* (**2**).
- Y. Aluma, Z. Baron, R. Barrett, C. Baumgartner, G. Beguš, S. Bhattacharya, M. M. Bronstein, S. Dahan, O. Davis, S. de Haas, J. Defoe, J. DelPreto, R. Dessì, R. Diamant, J. Gatesy, K. George, S. Gero, D. Gibbons, D. Gibbons, S. Gil, S. Goldwasser, D. F. Gruber, O. Harve, A. Hernandez, M. Ishay, N. Jadhav, L. KC, A. Kenny, A. Leitao, M. Lucas, A. Maalouf, P. Malkin, Y. Mevorach, S. Pagani, O. Paradise, G. Petri, S. Poetto, **E. Rossi**, D. Rus, M. Salino-Hugg, A. Santoro, P. Sharma, D. Tchernov, A. Torralba, P. Tønnesen, D. M. Vogt and R. J. Wood. 2026. Description of a collaborative sperm whale birth and shifts in coda vocal styles during key events. *Scientific Reports.* (**1**).
- V. Singh, M. Khanzadeh, V. Davis, H. Rush, **E. Rossi**, J. Shrader and P. Lio. 2025. Bayesian Binary Search. *Algorithms.* (**6**).

Peer-reviewed conference proceedings

- V. Davis, V. Singh and **E. Rossi**. 2025. Channel Balance Interpolation in the Lightning Network via Machine Learning. *IEEE International Conference on Blockchain and Cryptocurrency (ICBC).* (**4**).
- J. Gastinger, S. Huang, M. Galkin, E. Loghmani, A. Parviz, F. Poursafaei, J. Danovitch, **E. Rossi**, I. Koutis, H. Stuckenschmidt, R. Rabbany and G. Rabusseau. 2024. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. *Advances in Neural Information Processing Systems.* (**47**).
- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2024. Do We Really Need to

Drop Items with Missing Modalities in Multimodal Recommendation?. *Proceedings of the 33rd ACM International Conference on Information.* (**50**).

◦ S. Huang, F. Poursafaei, R. Rabbany, G. Rabusseau and **E. Rossi**. 2024. UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs. *Learning on Graphs Conference (LoG).* (**20**).

◦ S. Huang, F. Poursafaei, J. Danovitch, M. Fey, W. Hu, **E. Rossi**, J. Leskovec, M. M. Bronstein, G. Rabusseau and R. Rabbany. 2023. Temporal Graph Benchmark for Machine Learning on Temporal Graphs. *Advances in Neural Information Processing Systems.* (**273**).

◦ **E. Rossi**, B. Charpentier, F. Di Giovanni, F. Frasca, S. Günnemann and M. M. Bronstein. 2023. Edge Directionality Improves Learning on Heterophilic Graphs. *Learning on Graphs Conference (LoG).* (**202**).

◦ B. Chamberlain, S. Shirobokov, **E. Rossi**, F. Frasca, T. Markovich, N. Hammerla, M. M. Bronstein and M. Hansmire. 2022. Graph Neural Networks for Link Prediction with Subgraph Sketching. *International Conference on Learning Representations (ICLR).* **Oral (top 5%).** (**230**).

◦ M. Mutti, R. De Santi, **E. Rossi**, J. F. Calderon, M. M. Bronstein and M. Restelli. 2022. Provably Efficient Causal Model-Based Reinforcement Learning for Systematic Generalization. *Proceedings of the AAAI Conference on Artificial Intelligence.* (**36**).

◦ **E. Rossi**, F. Monti, Y. Leng, M. M. Bronstein and X. Dong. 2022. Learning to Infer Structures of Network Games. *Proceedings of the 39th International Conference on Machine Learning, ICML.* (**13**).

◦ **E. Rossi**, H. Kenlay, M. I. Gorinova, B. Chamberlain, X. Dong and M. M. Bronstein. 2022. On the Unreasonable Effectiveness of Feature propagation in Learning on Graphs with Missing Node Features. *Learning on Graphs Conference (LoG).* (**196**).

◦ B. Chamberlain, J. Rowbottom, M. I. Gorinova, S. D. Webb, **E. Rossi** and M. M. Bronstein. 2021. GRAND: Graph Neural Diffusion. *Proceedings of the 38th International Conference on Machine Learning, ICML.* (**544**).

◦ B. Chamberlain, **E. Rossi**, D. Shiebler, S. Sedhain and M. Bronstein. 2020. Tuning Word2vec for Large Scale Recommendation Systems. *RecSys - 14th ACM Conference on Recommender Systems.* (**38**).

Peer-reviewed workshop proceedings

◦ D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà and **E. Rossi**. 2025. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS Workshop on AI for Animal Communication.* **Spotlight (top 15%).** (**3**).

◦ J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, **E. Rossi**, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, H. Stärk, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede and L. Naef. 2024. PLINDER: The Protein-Ligand Interactions Dataset and Evaluation Resource. *ICML ML for Life and Material Science Workshop.* (**95**).

◦ **E. Rossi**, B. Chamberlain, F. Frasca, D. Eynard, F. Monti and M. Bronstein. 2020. Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML Workshop on Graph Representation Learning.* (**1747**).

◦ **E. Rossi**, F. Frasca, B. Chamberlain, D. Eynard, M. Bronstein and F. Monti. 2020. SIGN: Scalable Inception Graph Neural Networks. *ICML Workshop on Graph Representation Learning.* (**600**).

◦ **E. Rossi**, F. Monti, M. Bronstein and P. Lio. 2019. ncRNA Classification with Graph Convolutional Networks. *KDD Workshop on Deep Learning on Graphs.* (**40**).